

Lesson 17

Challenges in Dashboard Data

- 1. **Data Latency:** The delay between the event and the update in the dashboard.
- 2. **Data Inconsistency:** Conflicting sources and definitions of data.
- 3. **Missing Data:** Incomplete records or gaps in the data system.
- 4. **Overload:** Too much data leading to cognitive fatigue and difficulty in decision-making.

Granularity of Dashboards:

- **High-Level (Strategic):** Aggregated KPIs, quarterly trends.
- **Mid-Level (Tactical):** Weekly performance, campaign results.
- **Low-Level (Operational):** Real-time transactions, incident counts.

Tailor the **granularity** of the dashboard to the user's needs.

Data Context : Making Metrics Meaningful

- **Numbers Alone Aren’t Enough:** Show actual vs. target using benchmarks, goals, or past periods.
- **Tools to Provide Context:**
 - Reference lines in charts.
 - Status icons .
 - Bullet charts and gauges.

Insight: Context transforms numbers into meaningful narratives.

Criteria for Dashboards

Criteria	Strategic	Analytical	Operational	Tactical
Update Frequency	Periodic	On-demand	Real-time	Daily/weekly
Interactivity	Low	High	Medium	Medium
Audience	Executives	Analysts	Operators	Managers
Key Focus	KPIs/Trends	Exploration	Alerts/Status	Action Plans

Types of Dashboards

1. Strategic Dashboards

- **Purpose:** High-level monitoring of KPIs and long-term goals.
- **Users:** Executives and senior leaders.
- **Features:** Show overall business health; updated periodically (daily/weekly).
- **Example Widgets:** Revenue trend, profit margins, customer churn.

2. Analytical Dashboards

- **Purpose:** Enable exploration of data, uncover trends, and perform diagnostics.
- **Users:** Analysts, data scientists, business intelligence teams.
- **Features:** Interactive filters, drill-downs, complex visualizations (heat maps, scatter plots).
- **Common Uses:** Forecasting, root cause analysis, segmentation studies.

3. Operational Dashboards

- **Purpose:** Monitor real-time or near real-time operations.
- **Users:** Managers, supervisors, front-line teams.
- **Features:** Highly time-sensitive, auto-refreshing, alert-based or threshold-driven.
- **Example:** System outage alerts, order fulfillment status.

4. Tactical Dashboards (Hybrid)

- **Purpose:** Guide mid-term decision-making by combining strategic and operational elements.
- **Users:** Mid-level managers, department heads.
- **Features:** Blend of summary KPIs and detailed performance metrics, supporting short- to medium-term planning and adjustments.
- **Example:** Marketing campaign performance dashboard.

Dashboard for Usability

- **Organize the Information:** Support the meaning and use of data through visual clarity.
- **Consistency:** Use consistent colors and design elements for easy interpretation.
- **Aesthetically Pleasing:** Design with acceptance and accessibility in mind.
- **User-Driven Narrative:** Include drill-down and interactivity features.
- **Usability Testing:** Prototype design and test usability.

Dashboard Utilities

- **Low or No-Code Tools:**
 - Tableau
 - Power BI
 - Google Data Studio
- **Coding-Based Tools:**
 - Python
 - Dash (Plotly)
 - Voila (Jupyter notebooks)
 - R Packages
 - Shiny (R Studio)
 - Flexdashboard